



MULTIMEDIA UNIVERSITY OF KENYA

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

UNIVERSITY EXAMINATIONS 2017/2018

SECONDYEAR SECOND SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE IN COMPUTER TECHNOLOGY AND INFORMATION
TECHNOLOGY

ICS 2105: DATA STRUCTURES AND ALGORITHMS

DATE: WEDNESDAY 29TH AUGUST 2018



TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED.

ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS

QUESTION ONE (THIRTY MARKS)

- a) Explain two advantages of linked lists compared to array
- b) Distinguish between time and space complexity of an algorithm
- c) Explain the following tree properties, giving an example
 - i. Leaf
 - ii. Depth of a tree
 - iii. Degree of a node
 - iv. Complete tree

3 marks

4 marks

8 marks

- d) Explain the following
- i. Importance of algorithm analysis 4 marks
 - ii. Criteria for algorithm analysis 2 marks
- e) Explain the term linear data structure
- f) Describe stack operations 5 marks
- g) use selection sort to sort the following values in ascending order 4 marks
- 56, 20, 90, 23, 120, 77

QUESTION TWO (TWENTY MARKS)

- a) Explain bubble algorithm, giving an advantage and disadvantage 4 marks
- b) Illustrate bubble sort using the following values

40	5	25	70	15	3
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6 marks

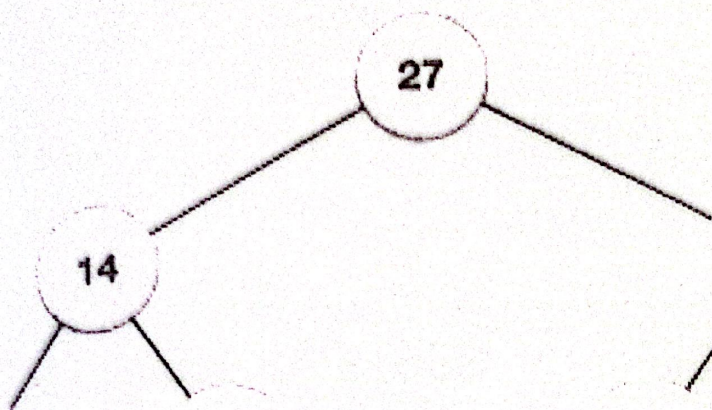
Write a C++ program which prompts the user to enter ten(10) values into an array and sort them in ascending order. The program should display the sorted values. 10 marks

QUESTION THREE (TWENTY MARKS)

- a) Write down the prefix and post fix notation of $a^b (c^k + d/e)$ 6 marks
- b) Find the value of P in the following postfix expression 4 marks
- $P = xya^b c - * +$ where $x=4, y=3, a=2, b=7, c=5$
- c) Generate a binary search tree using the following values 4 marks

50	25	45	60	80	70	35	55	65	40
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- d) Traverse the binary search tree below using all available traversal methods 6 marks



QUESTION FOUR (TWENTY MARKS)

a) Describe Heap data structure, giving a suitable example

2 marks

b) Use the following values to generate a Max-heap and then illustrate heap sort .

24, 56, 47, 35, 10, 90, 82, 31

10 marks

c) Suppose A,B and C are 2X3 matrices. Write a program which prompts the user to enter values into matrix A and matrix B and then calculate and store their sum in Matrix C. The program should also display the three matrices

8 marks

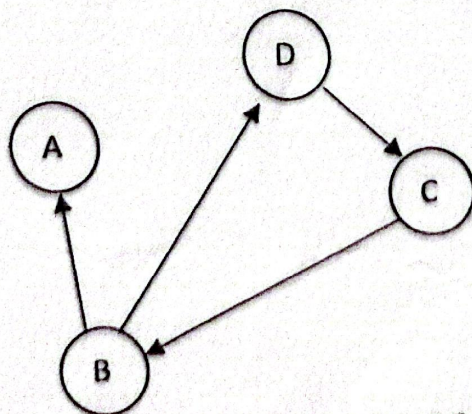
QUESTION FIVE (TWENTY MARKS)

a) Distinguish between a directed and undirected graph

4 marks

b) Identify Two commonly used representations of graphs and apply them to represent the graph below.

8 marks



c) Write a C ++ function to implement the POP and PUSH functions of stack ADT

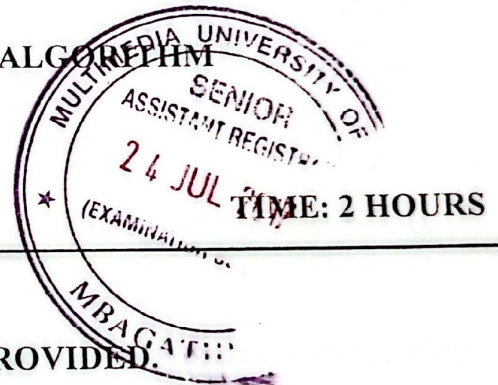
8 marks



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BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY AND COMPUTER
TECHNOLOGY

ICS 2105: DATA STRUCTURES AND ALGORITHMS

DATE: WENDSDAY 5TH JULY, 2017 JULY 2017



INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED.
ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS

QUESTION ONE (THIRTY MARKS)

- a) Define the following terms in reference to binary tree
- i. Leaf
 - ii. Depth of a tree
 - iii. Degree of a node

(6 Marks)

b.c.w

- b) Describe the advantages of a linked list compared to an array (4 Marks)
- c) Describe the most appropriate data structure in each of the following
- Map of the Puget Sound highway system used to display traffic travel times on a web page. The map displays principle cities, intersections, and major landmarks, the roads that connect them, and the travel times between them along those roads. Travel times along the same road may be different in different directions. (1 mark)
 - Chess board – an 8 x 8 board used for a game of chess. Each square on the board is either empty or contains a chess piece. (6 Marks)
- d) Describe the various methods of traversing a binary tree (4 Marks)
- e) Distinguish between full binary tree and complete binary tree, giving a suitable example. (6 Marks)
- f) Explain Three different types of algorithms (2 Marks)
- g) Convert following infix expression to postfix expression:
 $A^B(C+D * F) - G / H$ (4 Marks)

QUESTION TWO (TWENTY MARKS)

- a) Differentiate between a linear and non-linear data structure. (4 Marks)
- b) Describe binary search tree (4 Marks)
- c) Generate a binary search tree from the following values
- | | | | | | | | | | |
|----|----|----|----|----|----|----|----|----|----|
| 50 | 45 | 57 | 67 | 34 | 90 | 60 | 70 | 40 | 80 |
|----|----|----|----|----|----|----|----|----|----|
- d) Traverse the tree in 2c above using the following traversals (4 Marks)
- Pre-order traversal
 - Post-order traversal
 - In-order traversal
 - Level order traversal

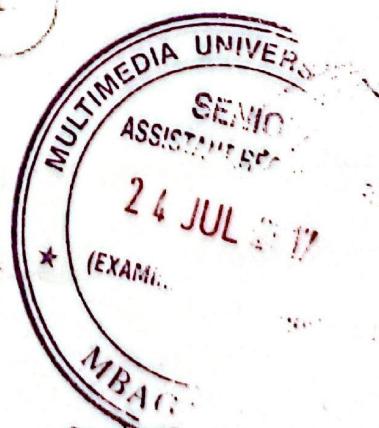
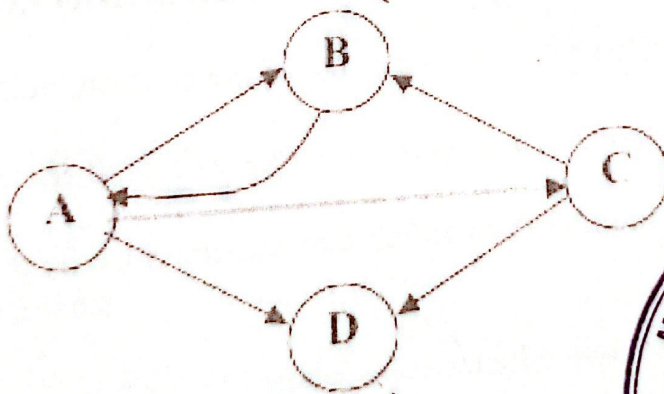
(8 Marks)

QUESTION THREE (TWENTY MARKS)

- Explain the different types of linked lists. (6 Marks)
- Distinguish between PUSH and POP (4 Marks)
 - using a program flowchart, design a program to PUSH items into a stack. (4 Marks)
- Write a C++ program to PUSH and POP items into a stack (6 Marks)

QUESTION FOUR (TWENTY MARKS)

- Discuss the importance of algorithm analysis (5 Marks)
- Describe the standard ways in which a graph can be traversed (5 Marks)
- Describe the term Minimum Spanning Trees, giving an example (5 Marks)
- Draw the adjacency matrix for the graph below (5 Marks)



QUESTION FIVE (TWENTY MARKS)

- Construct a min-heap using the following set of values

90	40	80	20	70	180	90	30	110	15
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(4 mark)

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b) Distinguish between space complexity and time complexity of an algorithm (3 Marks)

c) Illustrate bubble sort and selection sort using the following set of values in ascending order

210	40	70	25	150	56	85	30	90	10
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(6 Marks)

d) Write a C++ program which bubble sorts 10 values entered by the user in an array in descending order

(7 Marks)



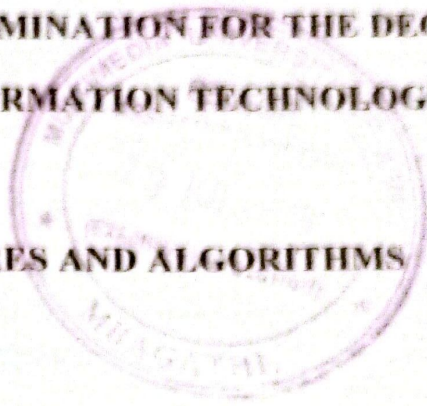
MULTIMEDIA UNIVERSITY OF KENYA

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

UNIVERSITY EXAMINATIONS 2015/2016

SECOND YEAR SECOND SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

ICS 2105: DATA STRUCTURES AND ALGORITHMS



DATE: TUESDAY 19TH APRIL 2016

TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED

ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS

QUESTION ONE (THIRTY MARKS)

1. Define the following terms.

[10 marks]

a. Data-structures

b. Algorithm

c. Linked list

d. Graph

e. Tree

2. List the operations that can be performed on a stack

[5 marks]

3. Write a code that implements the algorithm for enqueue operation.

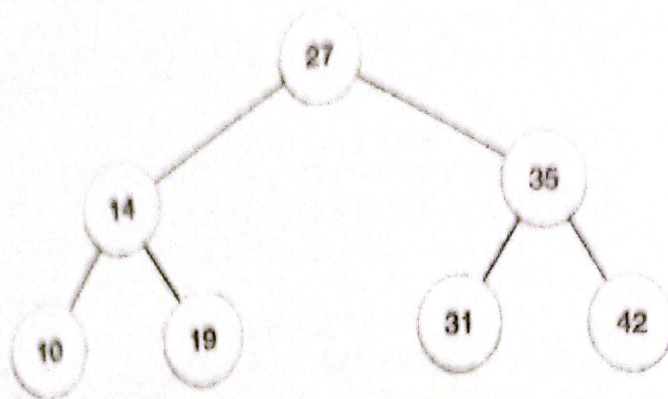
[5 marks]

4. Write a pseudocode for the linear search algorithm.

[5 marks]

5. See the below image of a binary search tree, and traverse it using all available methods –

[5 marks]



QUESTION TWO (TWENTY MARKS)

1. Write a code that demonstrates insertion operation at the beginning of a doubly linked list [5 marks]
2. Circular Linked List is a variation of Linked list in which first element points to last element and last element points to first element. Write a C program that demonstrates its implementation. [8 marks]
3. Define peek and write its code as implemented in stack. [2 marks]
4. Write an algorithm on evaluating postfix notation. [5 marks]

QUESTION THREE (TWENTY MARKS)

1. Binary search is a fast search algorithm with run-time complexity of $O(\log n)$. Write a pseudo code of binary search algorithm. [6 marks]
2. Interpolation search is an improved variant of binary search. Write a pseudo code to search 'target' data value index, using position probing. [8 marks]
3. Write a C program for the implementation of bubble sort [6 marks]

QUESTION FOUR (TWENTY MARKS)

1. Write a code that adds a vertex to a vertex list in a depth first search graph. [4 marks]
2. With the help of diagrams write steps for breadth first search. [6 marks]
3. Write a C program that implements breadth first search algorithm. [10 marks]

QUESTION FIVE (TWENTY MARKS)

1. Define the term traversal. [2 marks]
2. Quick sort is a highly efficient sorting algorithm and is based on partitioning of array of data into smaller arrays. Write the quick sort pseudo code. [5 marks]
3. Write a structure that represents a tree node. [3 marks]
4. Write an algorithm for insert operation in a tree. [5 marks]
5. With the help of a diagram, describe the post order tree traversal. [5 marks]



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UNIVERSITY EXAMINATIONS 2018/2019

SECOND YEAR FIRST SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND SOFTWARE ENGINEERING

CCS 2214: DATA STRUCTURES AND ALGORITHMS

DATE: TUESDAY 17TH DECEMBER, 2019

TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN THE ANSWER BOOKLET PROVIDED.

ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS

QUESTION ONE (THIRTY MARKS)

Describe binary search tree (3 Marks)

Describe the Graph data structure and its application in computing (5 Marks)

Explain any THREE queue operations (6 Marks)

Define an algorithm and highlight any five properties of a good algorithm (6 Marks)

Using a program flowchart design and write a C++ program to calculate the volume of a sphere. (6 Marks)

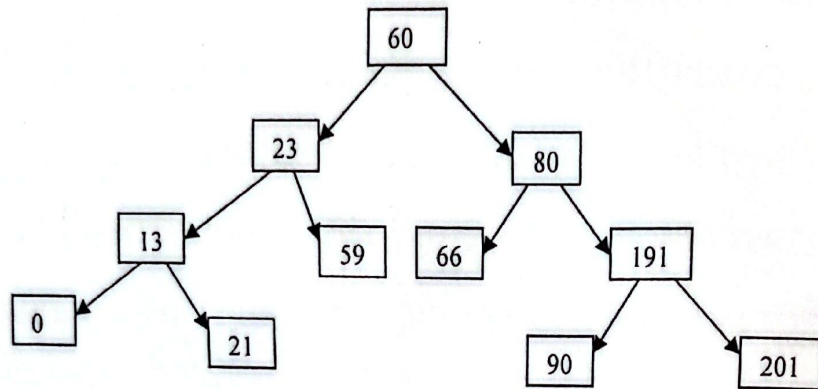
Illustrate selection sort using the following values, to sort in descending order

45	90	23	70	160	50	32	90	200	65
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(4 Marks)

QUESTION TWO (TWENTY MARKS)

a) Traverse the Binary tree below using the various traversal methods



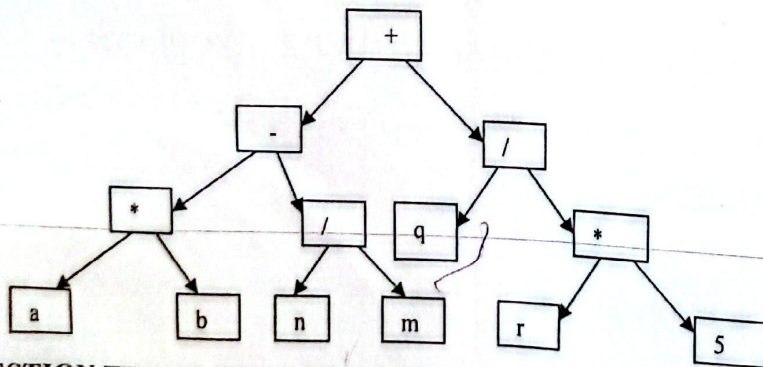
(8 Marks)

b) i) Convert the infix expression $(a*b-n/m) + q/r*5$ into corresponding prefix and postfix expression

(6 Marks)

ii) Represent the expression in b(i) above using an binary expression tree

(6 Marks)



QUESTION THREE (TWENTY MARKS)

a) Compare an array and a linked list giving an advantage and a disadvantage of each (5 Marks)

b) Explain the importance of the Big 'O' notation (3 Marks)

c) State and explain three examples of Big 'O' notation (6 Marks)

d) Using a program flowchart, illustrate the push() and pop() operations in a stack (6 Marks)

QUESTION FOUR (TWENTY MARKS)

Explain any THREE factors that should be considered when evaluating the efficiency of an algorithm. (6 Marks)

Illustrate bubble sort using the following data, to be sorted in ascending order

56, 34, 78, 150, 89, 23, . (5 Marks)

Write a program which prompts a user to enter TEN elements at random in an array and bubble sort them in ascending order. The program should display the elements before sorting and after sorting (9 Marks)

QUESTION FIVE (TWENTY MARKS)

Explain the properties of a Heap data structure (4 Marks)

Generate a max heap using the following values in the order indicated

25, 40, 70, 10, 57, 90, 50 (5 Marks)

Given the following data items in an array:

45	67	23	18	78	53	38	40
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i. outline the steps for finding value 78 in the array using binary search algorithm (5 Marks)

ii. An algorithm will find the optimal solution either in its best case, average case or worst case scenarios. Explain the three cases with reference to c(i) above (6 Marks)